

Milestone 3 Report – Machine Learning Model Development

1. Introduction

Milestone 3 focuses on developing machine learning models to forecast demand using the dataset prepared in earlier milestones. Forecasting future resource demand is important for planning infrastructure, optimizing resource allocation, and supporting better decision-making in cloud environments. In this stage, historical and engineered economic data were used to train predictive models and evaluate their performance.

2. Objective

The objective of this milestone is to build machine learning models capable of predicting future usage patterns. The work includes training models on historical data, evaluating their accuracy using standard metrics, and selecting the most suitable model for forecasting tasks. This stage provides the analytical foundation required for deploying the forecasting solution in the next milestone.

3. Dataset Description

The dataset used in this milestone is the feature engineered dataset created during Milestone 2. The file 'economic_data_feature_engineered.csv' contains processed features derived from the original dataset. Feature engineering prepared structured data suitable for machine learning models and improved prediction capability.

4. Data Preparation

The dataset was loaded using the Pandas library. Input features and the target variable were separated, and the dataset was divided into training and testing subsets. The training data was used to train the machine learning model, while the testing data was used to evaluate model performance.

5. Model Development

Machine learning models were trained using the prepared dataset. Algorithms such as regression models or ensemble methods can learn patterns from historical data and generate predictions. The training process helps the model understand relationships between variables.

6. Model Evaluation

The trained model was evaluated using Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). MAE measures the average difference between predicted and actual values, while RMSE measures the square root of the average squared differences between predictions and actual results.

7. Results and Discussion

The machine learning model was successfully trained and evaluated. The evaluation metrics provided insights into the accuracy of predictions. The results indicate that the model can capture patterns within the dataset and produce useful forecasts.

8. Conclusion

Milestone 3 completed the model development stage of the project. The dataset was prepared, models were trained, and evaluation metrics were calculated. The next milestone will focus on integrating the forecasting model into a practical deployment environment.